

Atin Sakkeer Hussain

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Summary

AI Researcher & Applied Systems Engineer specializing in multimodal Large Language Models and building advanced AI harnesses tailored towards multi-agent reasoning, long-horizon task orchestration, adaptive memory architectures, and large-scale model evaluation. Published researcher with proven experience translating foundational research into robust production systems, ranging from developing simulated user and evaluation harnesses for Google Gemini at Turing to architecting enterprise-grade AI systems with persistent memory and tool execution. Backed by strong mathematical and first-principles deep learning foundations.

Education

Arizona State University, Bachelor of Science in Data Science

Jan 2025 - Jul 2026

- Data Science Major specializing in the Computer Science Track

National University of Singapore, Bachelor of Computer Science

Aug 2020 - May 2024

- Computer Science and Mathematics Major with a Minor in Philosophy
- NUS Turing Program Student specialized in Artificial Intelligence

Publications & Conferences

ARTEMIS-DA: An Advanced Reasoning and Transformation Engine for Multi-Step Insight Synthesis in Data Analytics by Atin Sakkeer

Jan 2025

- This paper introduces ARTEMIS-DA, a novel framework for complex multi-step data analytics that orchestrates three specialized components - a Planner for task decomposition, a Coder for dynamic code generation and execution, and a Grapher for interpreting visualizations. Its key novelty is the integration of structured reasoning, automated data transformation, and visual insight synthesis within a unified LLM-driven analytical workflow.
- Links:
 - <https://arxiv.org/abs/2412.14146>

M2UGen: Multi-modal Music Understanding and Generation with the Power of Large Language Models by Atin Sakkeer Hussain, Shansong Liu, Chenshuo Sun, Ying Shan

Dec 2024

- This paper introduces MuMuLLaMA, a novel multi-modal music understanding and generation framework, along with a 167.69-hour annotated dataset spanning text, images, videos, and music. Its key novelty lies in connecting LLM-based multi-modal reasoning with pretrained music generation models to jointly support music understanding, text-to-music generation, prompt-based editing, and multi-modal music generation.
- Links:
 - <https://arxiv.org/abs/2412.06660>
 - https://crypto-code.github.io/MuMu-LLaMA_Demo/
 - <https://github.com/shansongliu/MuMu-LLaMA>

Large Language Models for Judicial Entity Extraction: A Comparative Study by Atin Sakkeer Hussain, Anu Thomas

Jul 2024

- This paper investigates the use of LLMs for domain-specific entity recognition in Indian case law, focusing on judicial entities such as courts, judges, lawyers, petitioners, and FIR numbers. Its key contribution is a comparative evaluation of LLaMA 3, Mistral, and Gemma for extracting structured judicial facts, with Mistral and Gemma achieving the strongest balance of precision and recall.
- **Links:**
 - <https://arxiv.org/abs/2407.05786>

Music Understanding LLaMA: Advancing Text-to-Music Generation with Question Answering and Captioning by Shansong Liu, Atin Sakkeer Hussain, Chenshuo Sun, Ying Shan

Apr 2024

- This paper introduces MU-LLaMA, a Music Understanding LLM for music question answering and caption generation, addressing the lack of open-ended music QA data through the MusicQA dataset and a novel QA-pair generation methodology.
- **Links:**
 - <https://arxiv.org/abs/2308.11276>
 - <https://ieeexplore.ieee.org/document/10447027>
 - <https://github.com/crypto-code/MU-LLaMA>
- Accepted for ICASSP 2024 & Presented at the Conference in COEX, Seoul, South Korea on 17 April 2024

Combining Variational Autoencoders & Generative Adversarial Networks to Improve Image Quality by Atin Sakkeer Hussain

Dec 2019

- This paper introduces a novel GAN-VAE framework for improving low-quality images generated by GANs, particularly under model collapse and limited-data settings. Its key novelty is using a VAE trained on both real and GAN-generated images to transform generated samples into higher-quality versions, while demonstrating improved performance over standard WGANs on sparse, diverse datasets.
- **Links:**
 - <https://osf.io/preprints/osf/8bmdu>

Work Experience

AI Engineer and Python Developer at Turing

Jun 2024 - Aug 2026

- Developed a User Simulation framework to train and evaluate Gemini flagship models with realistic user interaction scenarios using LLM Simulated Users in a real-world environment with dummy APIs.
- Designed and implemented scalable AI frameworks for training data creation and automated quality control, enabling high-quality dataset generation and validation for Gemini model training.
- Developed comprehensive evaluation frameworks for multi-turn conversational AI scenarios, incorporating simulated user interactions to assess system performance and behavior under realistic dialogue conditions - inspired by advanced benchmarks such as Tau Bench and Galileo AI.
- Collaborated closely with Google's Gemini team to support large-scale data creation and evaluation for training and benchmarking AI models across diverse, real-world use cases, ensuring data integrity, relevance, and alignment with evolving model objectives.

- Built evaluation framework for other Google products powered by Gemini models for continuous improvement loops based on user feedback using common error classification and reinforcement learning.

AI Integration Specialist at Saniservice, Dubai, UAE

Jun 2024 - Aug 2026

- Integrated and optimized automated data pipelines into the company's daily workflows, incorporating advanced image classification, image analysis, and automated report generation modules. The system automated key stages of data processing and transformed raw visual and operational data into structured, actionable reports, reducing manual effort and processing time for delivering work completion reports to clients. This streamlined routine workflows, improved data consistency and accuracy, and significantly enhanced overall operational efficiency and turnaround time.
- Built and deployed an AI-human collaborative sales interface where personalized AI agents communicate with clients and prospects on behalf of sales representatives, handling routine interactions and seamlessly handing off qualified leads and final closing to the sales team. A custom memory framework enables agents to learn each salesperson's communication style and preferences while retaining client-specific context, allowing the AI to step in when sales representatives are unavailable and maintain continuity across interactions.
- Architected, developed, and deployed a custom AI framework that seamlessly connected with the company's ERP system, WhatsApp communication channels, and email servers, enabling a unified, intelligent chat interface. This system empowered staff to query, retrieve, and update operational data in real time through natural language interactions, enhancing communication, accessibility, and decision-making across departments.

Undergraduate Researcher at NUS, Singapore

Jun 2023 - May 2024

- Conducted research in Natural Language Processing and AI-driven Music Understanding and Generation, investigating multimodal approaches that connect music, language, and machine learning. Explored tasks including Music Question Answering, music representation learning, and multimodal music-language modeling to enable AI to understand, reason about, and generate music through natural language.
- Developed and published two novel models, MU-LLaMA and M2UGen, advancing state-of-the-art approaches in AI-driven music understanding and generation. The work focused on improving music representation learning, multimodal understanding, and generative capabilities, enabling more effective interaction between language models and musical content.

Undergraduate Researcher at Ubisoft, Singapore

Jan 2022 - Dec 2023

- Investigated deep learning architectures for recommendation systems, focusing on Transformer-based models designed to capture user behavior, preferences, and sequential interactions for personalized recommendations in gaming environments.
- Proposed and implemented a novel session-based recommendation framework that combined user preference clustering with sequential modeling to generate more personalized recommendations. Evaluated the approach on real-world and proprietary Ubisoft datasets, demonstrating strong performance and its potential to enable more adaptive, context-aware in-game recommendation systems.

AI Integration Specialist at DLI-IT Group HR Works, Dubai, UAE

Jun 2022 - Aug 2022

- Designed and deployed an intelligent HR chatbot that automated internal employee service requests, provided instant access to HR information, and streamlined routine HR workflows, improving response efficiency and reducing manual workload.
- Built an AI-powered resume parsing and matching system that automatically extracted key candidate information and evaluated it against job descriptions, enabling faster candidate screening and significantly accelerating recruitment workflows.
- Trained and mentored HR personnel to independently maintain, enhance, and adapt both AI systems, enabling continuous improvement and ensuring the solutions remained aligned with evolving organizational requirements.

Computer Vision Engineer on Project Cyclops with FarmBot, Duabi, UAE

Jun 2021 - Aug 2021

- Developed and deployed a precision laser weeding attachment for FarmBot, integrating automated laser-based targeting to selectively eliminate weeds while minimizing the need for chemical herbicides and enabling more sustainable agricultural practices.
- Engineered a light-weight computer vision model for real-time weed detection and targeting, integrating visual recognition with automated positioning algorithms to accurately identify and target weeds. The system improved targeting precision and operational efficiency, demonstrating the potential of AI-driven automation for precision agriculture.

AI Trainer at Sustainable Living Labs, Singapore

May 2021 - Jul 2022

- Delivered AI education to secondary and polytechnic students through Intel's AI for Youth Program, introducing core AI and machine learning concepts and mentoring student teams through the development of socially impactful, real-world AI projects.
- Led Meta's AI Bootcamp at Temasek Polytechnic, teaching applied machine learning concepts and guiding students through the end-to-end development of AI projects, from problem formulation and model development to implementation and presentation.

Open-Source Projects

Stock Market AI Prediction & Automated Virtual Trading Platform

Feb 2020 - Jul 2020

- Built an end-to-end LSTM time-series forecasting engine with technical indicator feature engineering, paired with an automated trading bot and interactive Flask web UI (460+ GitHub stars, 100+ forks).
- Implemented multi-step recurrent sequence modeling to minimize mean squared forecasting error across multi-year historical equity data.

NeuroEvolution (NEAT) & Reinforcement Learning Agents

Aug 2023 - Dec 2023

- Applied NeuroEvolution of Augmenting Topologies (NEAT) and genetic algorithms to autonomously evolve neural topologies that cleared levels in Super Mario World and Chrome T-Rex runner.
- Built Actor-Critic deep reinforcement learning agents to play arcade environments in the Atari Gym.

Foundational Neural Architectures & Associative Memory from Scratch

Apr 2020 - May 2020

- Engineered 6+ core deep learning architectures (MLP, RNN, LSTM, GRU, GAN, Denoising Autoencoders) from first principles using pure NumPy matrix calculus and analytical backpropagation.
- Used these samples during my work as an AI Trainer to teach the fundamentals of deep learning methods without the abstraction offered by python libraries like PyTorch.
- Implemented Hopfield recurrent networks leveraging energy minimization dynamics for associative memory retrieval and pattern reconstruction from noisy matrices.

Open-Source Code for MU-LLaMA and M²UGen

Aug 2023 - Dec 2023

- Built evaluation pipelines and interactive demonstration interfaces for multimodal music LLMs (MU-LLaMA, M2UGen) enabling text-, video-, and audio-to-music cross-modal generation.
- Benchmarked downstream tagging performance across diverse audio foundation model representations on datasets such as the MagnaTagATune dataset.

Generative Modeling, Latent Manifolds & Neural Art

Oct 2019 - Jan 2020

- Built Autoencoders and VAEs to project high-dimensional data into continuous 2D latent manifolds, implementing vector interpolation for smooth generative morphing.
- Implemented neural style transfer with Gram matrix loss optimization and DeepDream algorithmic pareidolia via gradient ascent on intermediate VGG19 feature activations.

Sequence-to-Sequence NLP & Language Understanding

Sep 2019 - Jan 2020

- Built Encoder-Decoder translation pipelines with Bahdanau additive attention mechanisms and developed recurrent GRU dialogue agents.
- Evaluated Transformer (BERT/SciBERT) and BiLSTM-CRF architectures for scientific keyphrase extraction and classification on the ScienceIE benchmark.

References

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